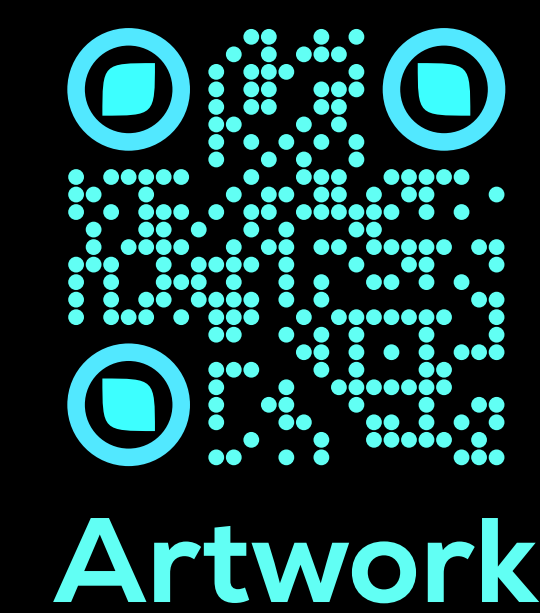




Abstract

Sensorium Arc (AI reflects on climate) is a real-time multimodal interactive AI agent system that personifies the **Ocean as a poetic speaker** and guides users through immersive explorations of marine data.

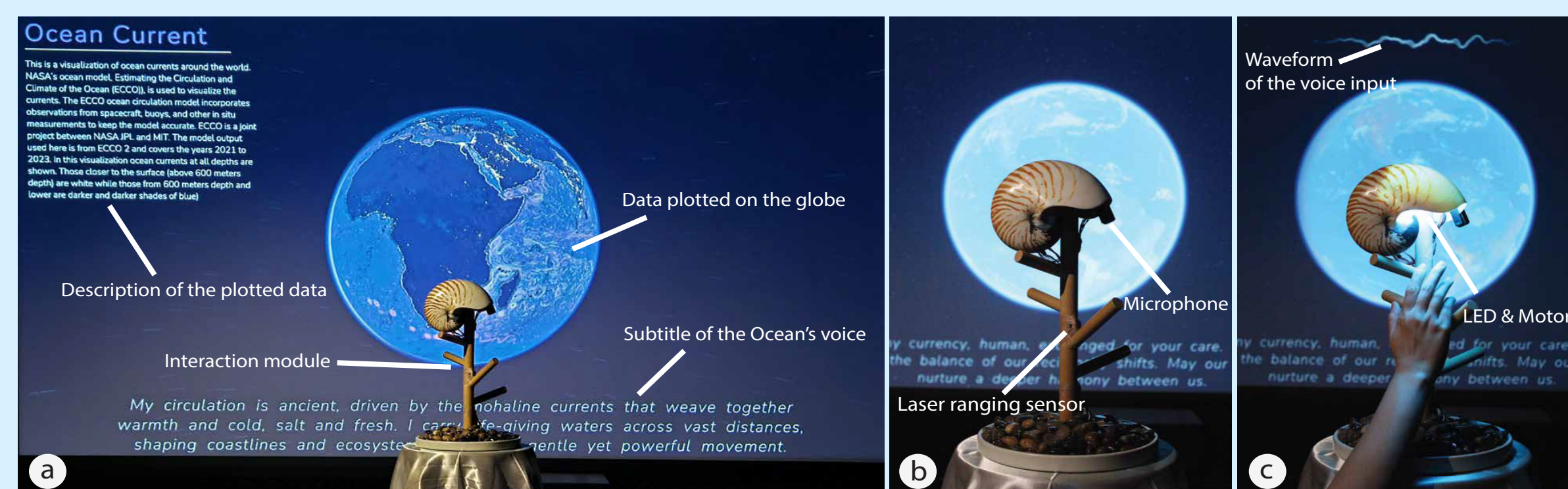
Built on a **modular multi-agent system** and retrieval-augmented large language model (LLM) framework, Sensorium enables natural spoken conversations with an AI that embodies the ocean's perspective, blending scientific insight with ecological poetics. Through keyword detection and semantic parsing, the system triggers data visualizations and audio-visual playback based on time, location, and thematic cues drawn from the dialogue. Inspired by the eco-aesthetic philosophy of Newton Harrison, Sensorium Arc reimagines **ocean data not as abstraction but as living narrative**. The project demonstrates the potential of conversational AI agents to mediate intuitive, affective access to high-dimensional environmental data and proposes a new paradigm for **human-machine ecosystem interaction**.

How to Talk to Sensorium

Sensorium Arc can hold emotional or conversational dialogue, interpret location-specific questions, and respond to deeper scientific inquiries that explore interconnected oceanic and climatic processes.

To speak with Sensorium:

1. Reflect on what you would like to ask the ocean
2. Approach Sensorium's earpiece, the nautilus shell, moving your hands closer until it illuminates with **blue ripples**, as if whispering into a seashell (within about 50cm)
3. Ask your question and move your hand away from the nautilus



Example Questions:

Casual: "Ocean Ocean, how are you feeling today?" - "What has been on your mind?"

Local: "How have California's coastlines changed in the past generation?"

Deep: "I've heard the ocean is dead in some areas, what does that actually mean?"

"Help me understand by relating it to human social issues."

Feel free to move beyond these examples and ask the ocean whatever questions feel most natural or important to you.

**Note Sensorium Arc is currently available only as a live, creator-hosted demonstration and is not publicly distributed in any standalone form. All spoken audio and conversation content are processed only on the local machine and are not stored or transmitted.

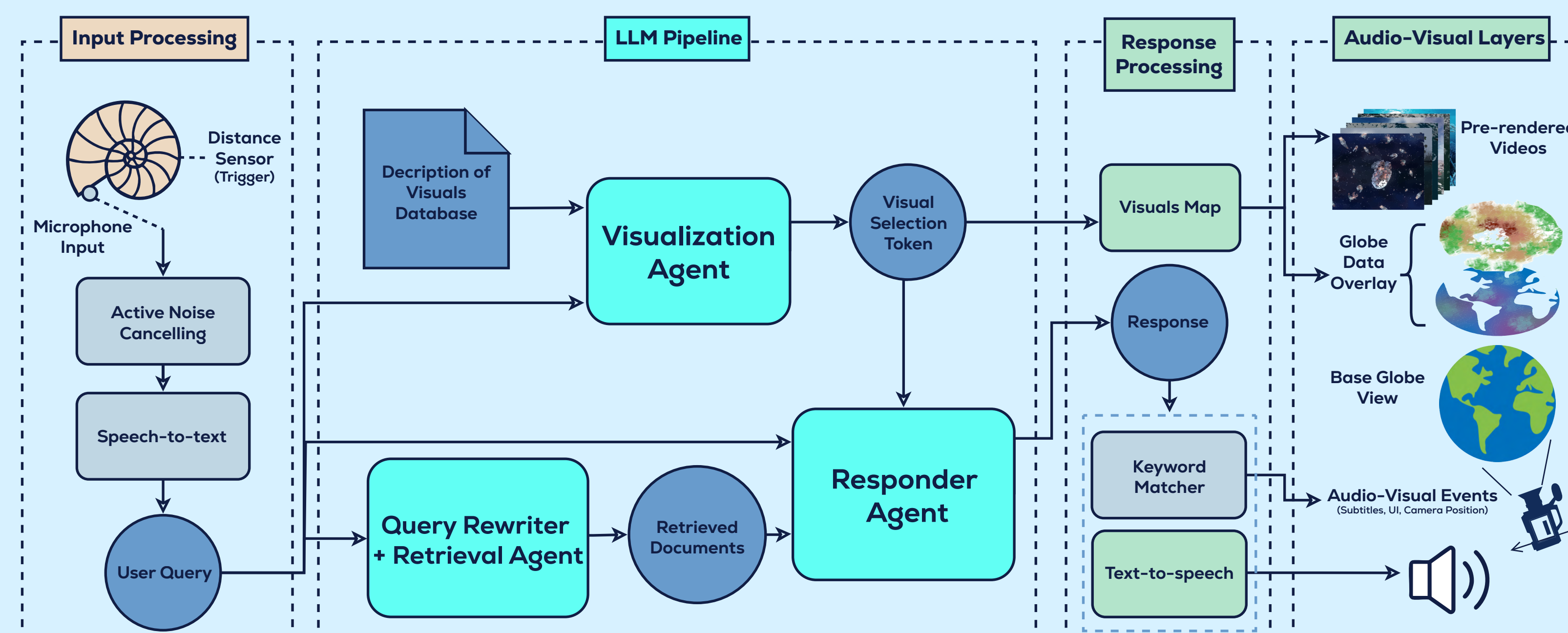
Architecture

The overall structure of the proposed system is modularized into

Input Processing, LLM Pipeline, Response Processing, and Audio-visual Layers.

1. Input Processing
 - i. Nautilus shell microphone housing: Arduino + distance sensor activation
 - ii. Voice to Text: Whisper (Unity Sentis runtime, on-device inference)
2. Multi-Agent LLM Pipeline
 - i. Query Rewriter Agent: Refines user queries for meaningful retrieval
 - ii. Retrieval Agent: Pulls from scientific data sets + curated eco-art corpus
 - iii. Visualization Decider Agent: Selects Relevant data visuals (e.g., sea surface temperature, chlorophyll, carbon dioxide emissions)
 - iv. Responder Agent: Generates poetic, ecologically grounded "Ocean Voice."
3. Response Processing
 - i. Keyword Matching: Maps tokens and keywords in LLM output to the audio-visual layer systems
 - ii. Text to Voice: Jets (Unity Sentis runtime, on-device inference)

Diagram of Architecture



Technical Implementation

Sensorium Arc is built in Unity 6 and integrates a custom Arduino-based Nautilus interface containing a distance sensor, LEDs, a microphone, a motor, and active noise cancellation. Speech input is transcribed through a Whisper STT pipeline.

The system uses a multi-model LLM stack within a modified LLMUnity framework:

- Llama-3.2-3B (Visualization Decider)
- Qwen 8B (Query Rewriter)
- Gemma 3 12B (Responder)

Retrieval is handled via **all-MiniLM-L12-v2 embeddings** with usearch ANN (k=2). Text-to-speech runs on the Jets TTS model through Unity Sentis. Scientific data layers (CO₂, SST, chlorophyll, currents, hyperspectral Kd) are preprocessed into globe-ready textures, with additional thematic videos rendered offline.

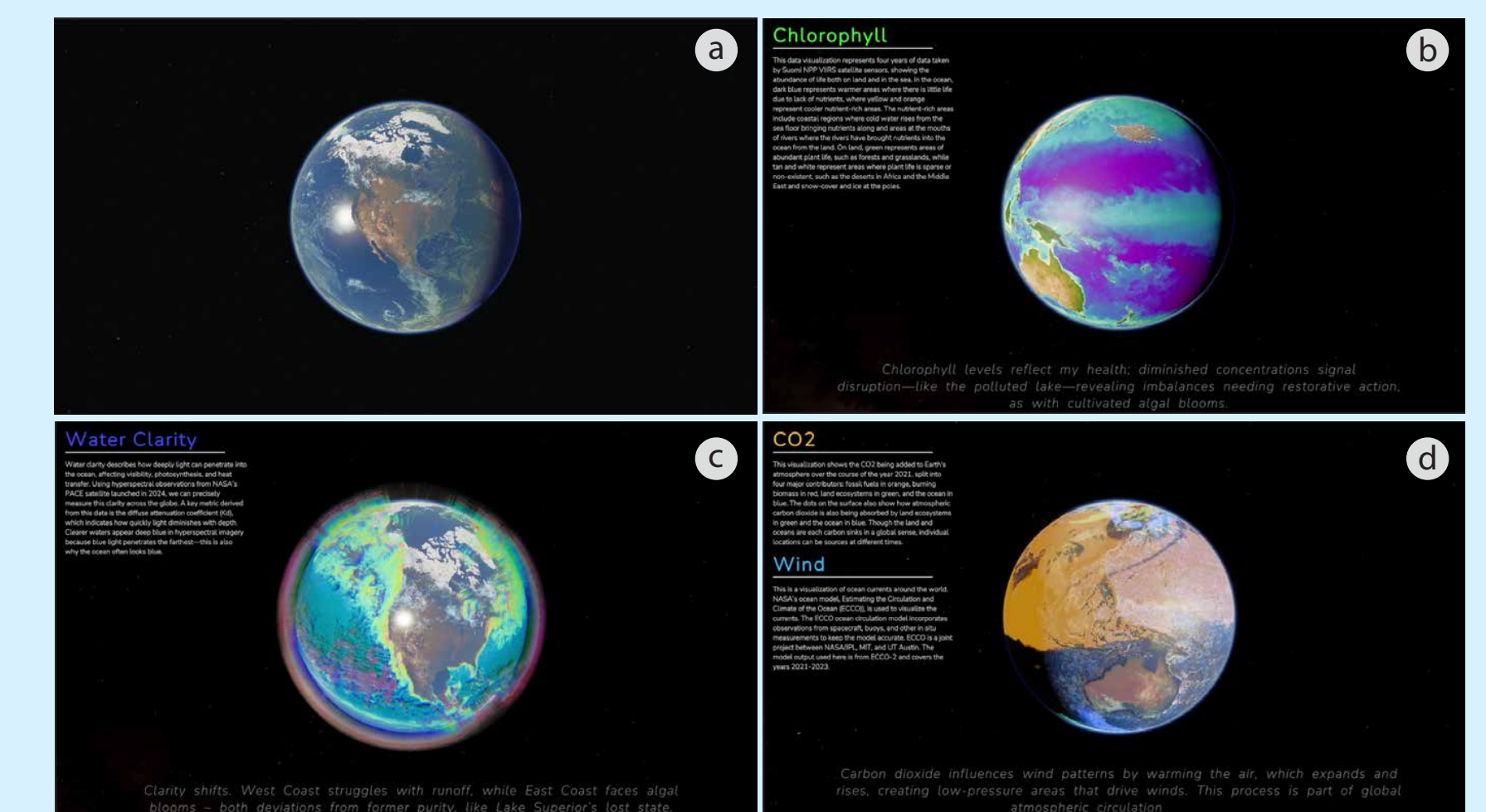
Development and deployment were performed on Windows 11, i9-13900HX, 64 GB RAM, and an RTX 4090 Laptop GPU, with CUDA acceleration for inference; macOS testing used Vulkan.

Visual Layers

Audio-visual layers are directly tied to the user's interactive experience, combining **data visualization, event triggers, video playback, localization, and text-to-speech (TTS)** audio accompanied by synchronized subtitles for accessibility.

Our **modular globe visualization database** include sea surface temperature, ocean currents, atmospheric CO₂ emissions, chlorophyll concentration, water clarity and more from diverse sources including NASA Earthdata and earthaccess.

Narrative-focused videos courtesy of Virtual Planet visualize sea-level rise, plastic waste dispersion, ocean acidification, and overfishing.



Screenshots of the different exemplary scenarios
(a) Default globe view before user interaction.
(b) Chlorophyll concentration data visualized on the globe surface.
(c) Water clarity rendered using layers of hyperspectral diffuse attenuation coefficient (Kd) from NASA's PACE satellite (2024).
(d) Combined visualization of atmospheric CO₂ levels and ocean surface wind flow, highlighting interconnected environment

Discussion & Future Works

Human-AI interaction in Sensorium Arc probes the changing role of humanity in technologically mediated ecological discourse. Participants enter an embodied dialogue with a machine that speaks as the Ocean, bringing human empathy, moral reasoning, and cultural memory into contact with AI strengths in data retrieval, multimodal synthesis, and a consistent ecological persona. Sustainability appears not only as a technical or environmental metric, but as a shared cultural practice co-authored with non-human agents. The work invites reflection on how human values and ethics are encoded in AI systems, and how such systems may reshape collective ecological imagination and responsibility.

The current implementation delivers an immersive ocean conversational experience but also reveals four key directions for refinement: improving RAG reliability, optimizing LLM agent performance, advancing artistic design, and expanding scientific and social impact. Future work will **diversify query rewriting and document typing, reduce latency by better balancing model size and deployment, and refine multi-agent coordination**. **Artistically**, we plan richer narrative visualization and sound, moving from global ocean views to local, explorable scenes. **Scientifically and socially**, we aim to run controlled user studies on learning, empathy, and behavior change, link emotional resonance to civic engagement, and integrate more diverse, multilingual marine datasets through open APIs.

Acknowledgement

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